

Laurent Colbois

ML/COMPUTER VISION ENGINEER · PHD IN MACHINE LEARNING & BIOMETRICS

Rue du Rhône 2, 1920 Martigny, Switzerland

+41 79 942 29 61 | laurent.colbois@gmail.com | May 2nd, 1996 | lcolbois | 0009-0006-3547-5841

Laurent Colbois



“Building robust ML systems with reproducible workflows and clean code.”

Summary

Computer Vision and Machine Learning Engineer with a PhD from UNIL/Idiap, specializing in production-ready ML systems and reproducible workflows. Strong expertise in MLOps practices including experiment tracking, containerization, and workflow orchestration. Experienced in developing and benchmarking deep learning models (CNNs, GANs, Vision-Language Models) with a focus on code quality and reproducibility. Seeking to apply technical expertise to build robust, scalable ML solutions in a collaborative engineering environment.

Work Experience

Idiap Research Institute

Martigny, Switzerland

POSTDOCTORAL RESEARCHER - VISION-LANGUAGE MODELS FOR BIOMETRICS

Apr. 2025 - Mar. 2026

- Implemented end-to-end benchmarking pipeline for Vision-Language Models (Gemma3, Qwen2.5-VL, InternVL3) using PyTorch, Huggingface Transformers, and vLLM for efficient inference at scale
- Built reproducible ML workflows with Hydra for configuration management, Weights & Biases for experiment tracking, and Snakemake for pipeline orchestration
- Developed novel evaluation metrics and automated analysis tools using Python, Pandas, and visualization libraries to quantify model performance across multiple dimensions
- Coordinated cross-functional team (ML researchers, software developers, legal staff) to deliver a forensic data collection platform, managing requirements, specifications, and technical constraints
- Won first place at a 9-day startup hackathon by building a real-time MIDI music generation system in Python, demonstrating rapid prototyping and product demo skills

Idiap Research Institute

Martigny, Switzerland

PHD RESEARCHER - COMPUTER VISION & DEEP LEARNING

Jan. 2021 - Mar. 2025

- Developed CNN-based detection systems for AI-generated images using PyTorch, implementing custom architectures and training pipelines with GPU acceleration on HPC clusters (Slurm)
- Conducted large-scale evaluation of face recognition systems against deepfake attacks, processing datasets of 100K+ images with automated benchmarking pipelines
- Published and maintained open-source research code on GitLab, emphasizing reproducibility through containerization (Apptainer/Singularity) and environment management (Conda, Pip)
- Mentored four master's students on ML projects, providing technical guidance on model training, evaluation, and code best practices
- Delivered technical presentations on AI and deepfakes to diverse audiences, translating complex ML concepts into accessible explanations

Idiap Research Institute

Martigny, Switzerland

RESEARCH INTERN

Jul. - Dec. 2020

- Created a synthetic face recognition dataset using StyleGAN2, implementing data generation pipelines and quality control automation
- Benchmarked multiple face recognition models (ArcFace, CosFace) on synthetic and real datasets, analyzing performance trade-offs

Logitech

Lausanne, Switzerland

RESEARCH INTERN - AUDIO MACHINE LEARNING

Feb. - Aug. 2019

- Built an audio scene classification model using CNNs and mel-spectrograms for automatic speaker equalization
- Deployed model to embedded hardware (microcontroller), optimizing for real-time inference with resource constraints
- Presented prototypes to cross-functional teams and delivered a working demo showcasing model integration

Education

1 Publications

2026	Conference Paper (Under Review) , Benchmarking the Explanatory Quality of Open-Weight Vision-Language Models in Face Recognition	Face and Gesture
2024	Conference Paper , Evaluating the effectiveness of attack-agnostic features for morphing attack detection	IJCB
2023	Conference Paper , Approximating Optimal Morphing Attacks using Template Inversion	IJCB
2022	Conference Paper , On the detection of morphing attacks generated by GANs	BIOSIG
2022	Conference Paper , Are GAN-based morphs threatening face recognition?	ICASSP
2021	Conference Paper , On the use of automatically generated synthetic image datasets for benchmarking face recognition	IJCB

Skills

Domains	Computer vision, Deep learning, Vision-Language Models, Biometrics, Audio processing, Reproducible research, Cross-functional communication
ML & Programming	Python, PyTorch, Huggingface Transformers, vLLM, Scikit-Learn, Pandas, NumPy, C++ (basic)
MLOps & DevOps	Git, Docker, Apptainer/Singularity, Conda, Pip, Pixi, Snakemake, Prefect, Weights & Biases, MLFlow, Hydra, Slurm, Unix/Linux
Languages	French (Mother tongue), English (Professional proficiency), German (B2)